

Azure Hosting Pricing Comparison

Gulf Craft AI Costing Assistant

Pricing estimates as of April 2026. Azure prices are in USD/month (East US or West Europe region).
Approximate AED conversion at 3.67 USD = 1 AED is used where noted.

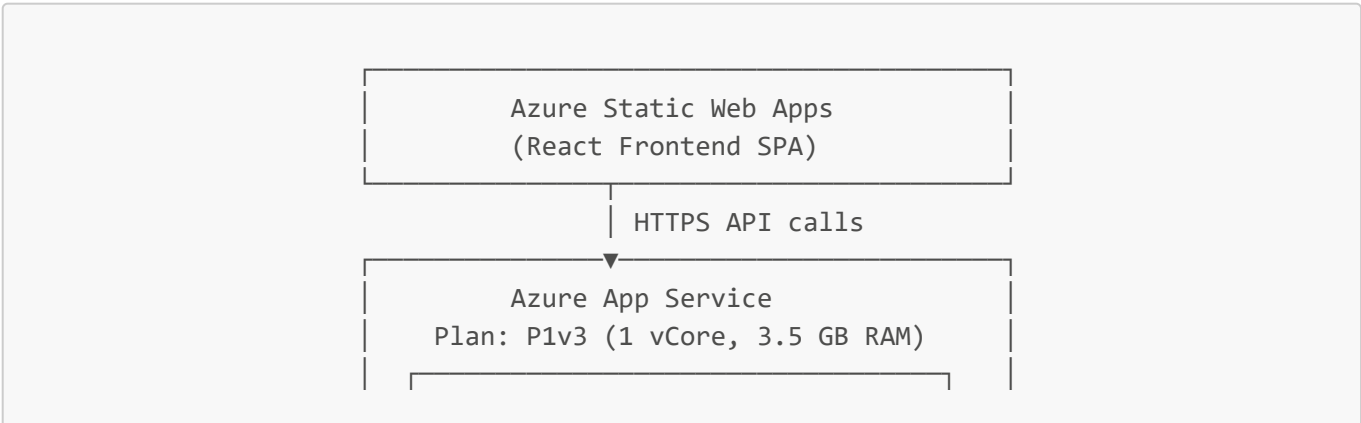
System Components to Host

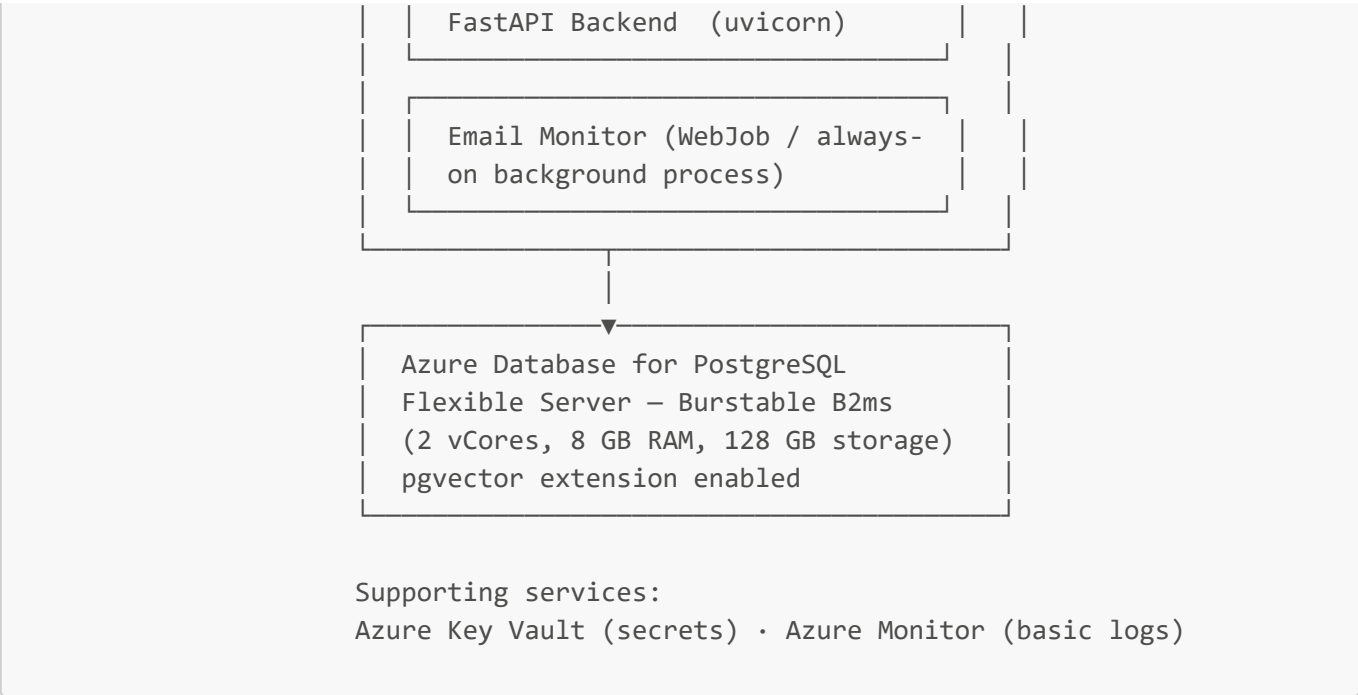
Component	Technology	Notes
Backend API	FastAPI (Python)	12-agent LangGraph orchestration
Frontend	React 18 / Vite static build	SPA
Database	PostgreSQL 15+ with pgvector	Vector search + relational data
Email Monitor	Python IMAP polling service	Background long-running process
Azure OpenAI	GPT-4.1, text-embedding-3-small, gpt-4.1-mini	External service (shared across both options)
Mistral OCR	Azure-hosted Mistral AI	External service (shared across both options)
Microsoft Graph	Mail + SharePoint	External API (no hosting cost)

Option 1 — Simple Barebone Deployment (Azure App Service)

Target use case: Internal team tool, low-to-medium concurrency (<20 simultaneous users), dev/staging environments, or early production with predictable load.

Design principles: Minimal managed services, PaaS-first, no container orchestration, single region, no redundancy.





Option 1 — Monthly Cost Breakdown

Infrastructure Services

Service	SKU / Tier	Spec	Est. USD/mo
Azure App Service Plan	Premium v3 P1v3	1 vCore, 3.5 GB RAM	\$73
Azure Static Web Apps	Standard	Global CDN, custom domain, auth	\$9
Azure Database for PostgreSQL Flexible Server	Burstable B2ms	2 vCores, 8 GB RAM	\$54
PostgreSQL Storage	128 GB (+ 7-day backup)	Includes pgvector	\$18
Azure Key Vault	Standard	Secrets, certificates	\$5
Azure Monitor / App Insights	Pay-as-you-go	5 GB log ingestion/month	\$10
Bandwidth (Egress)	East US outbound	~50 GB/month estimated	\$4
Total Infrastructure			~\$173/mo

Note: The App Service P1v3 runs both the FastAPI backend and the Email Monitor WebJob. If stability is a concern, upgrading to P2v3 (2 vCores, 7 GB, +\$73/mo) provides headroom.

AI/LLM Services (Variable — shared with Option 2)

These costs depend entirely on usage volume and are the same regardless of hosting tier.

Service	Pricing Basis	Low Usage	Medium Usage	High Usage
Azure OpenAI GPT-4.1 (chat agent)	\$15/1M input · \$60/1M output tokens	\$80	\$250	\$600
Azure OpenAI GPT-4.1-mini (matcher)	\$0.40/1M input · \$1.60/1M output tokens	\$5	\$20	\$60
Azure OpenAI text-embedding-3-small	\$0.02/1M tokens	\$2	\$8	\$25
Mistral AI OCR (Azure Marketplace)	Per-page pricing	\$10	\$40	\$120
AI/LLM Subtotal		~\$97/mo	~\$318/mo	~\$805/mo

Low: ~50 costing jobs/day · Medium: ~150 jobs/day · High: ~400+ jobs/day

Option 1 — Total Monthly Cost

Scenario	Infrastructure	AI/LLM	Total	Total (AED)
Low usage	\$173	\$97	~\$270/mo	~990 AED/mo
Medium usage	\$173	\$318	~\$491/mo	~1,802 AED/mo
High usage	\$173	\$805	~\$978/mo	~3,589 AED/mo

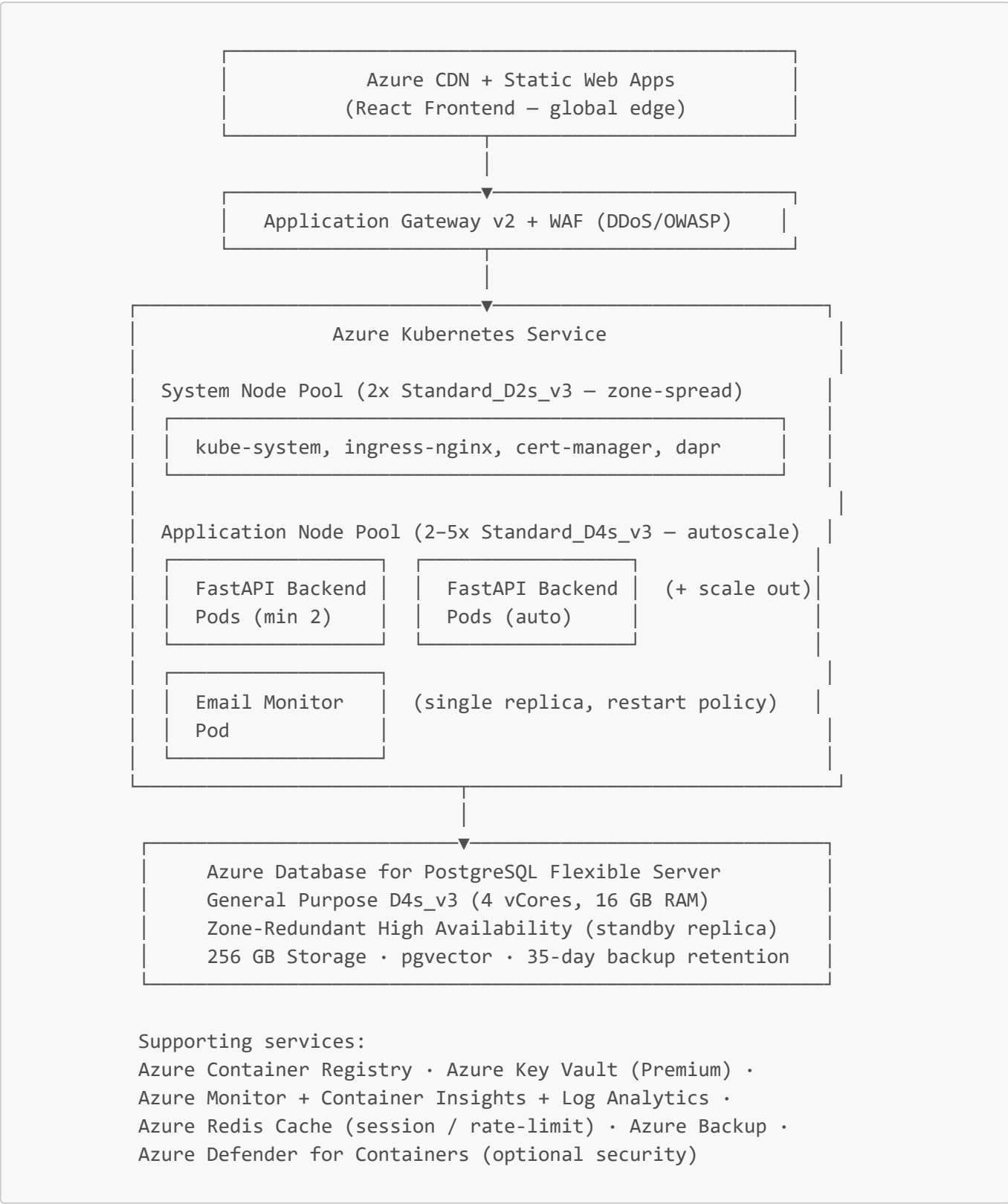
Option 1 — Limitations

Area	Limitation
Availability	Single instance, ~99.95% SLA. Any deployment restarts the service.
Scalability	Manual scale-up only. No auto-scaling on load spikes.
Database	Burstable compute — CPU credits can exhaust under sustained load. No standby/HA.
Email Monitor	Runs as WebJob; if App Service restarts, monitor restarts too (brief gap in IMAP polling).
Zero-downtime deploys	Not supported without deployment slots (requires Standard+ plan, +\$73/mo).
pgvector performance	8 GB RAM is sufficient for moderate vector search; degrades under large embedding tables.

Option 2 — Production-Ready AKS Deployment

Target use case: Full production workload, high availability requirement, enterprise SLA, auto-scaling under load spikes, fault tolerance, and operational observability.

Design principles: AKS orchestration, zone-redundant PostgreSQL, WAF-protected ingress, auto-scaling node pools, full observability stack, automated failover.



Option 2 — Monthly Cost Breakdown

AKS Cluster

Resource	SKU / Config	Spec	Est. USD/mo
AKS Management	Free tier	Control plane	\$0
System Node Pool	2× Standard_D2s_v3	2 vCores, 8 GB RAM each, zone-redundant	\$138
App Node Pool (avg 3 nodes)	3× Standard_D4s_v3	4 vCores, 16 GB RAM each, autoscale 2–5	\$414
AKS Load Balancer	Standard SKU	Included with node provisioning	\$18
AKS Cluster subtotal			~\$570/mo

Autoscaler triggers on CPU >70% or memory >80%. Pod disruption budgets ensure zero-downtime rolling deploys.

Data & Storage

Service	SKU / Tier	Spec	Est. USD/mo
Azure Database for PostgreSQL	General Purpose D4s_v3	4 vCores, 16 GB RAM	\$246
PostgreSQL Zone-Redundant HA	Standby replica, auto-failover <120s	Doubles compute cost	\$246
PostgreSQL Storage	256 GB + 35-day backup	pgvector, WAL archiving	\$47
Azure Redis Cache	C1 Standard (1 GB)	Session state, rate limits	\$55
Azure Backup (DB snapshots)	LRS vault	Daily + weekly retention	\$20
Data subtotal			~\$614/mo

Networking & Security

Service	SKU / Tier	Spec	Est. USD/mo
Application Gateway v2 + WAF	WAF_v2	OWASP rules, TLS termination, 2 capacity units	\$126
Azure CDN (frontend)	Standard Microsoft	100 GB egress/month	\$8

Service	SKU / Tier	Spec	Est. USD/mo
Bandwidth (Egress)	East US outbound	~150 GB/month (API traffic)	\$12
Azure Key Vault Premium	HSM-backed secrets	Keys, secrets, certs	\$10
Networking subtotal			~\$156/mo

Containers & CI/CD

Service	SKU / Tier	Spec	Est. USD/mo
Azure Container Registry	Standard	100 GB storage, geo-replication disabled	\$20
ACR subtotal			~\$20/mo

Observability

Service	SKU / Tier	Spec	Est. USD/mo
Azure Monitor + Application Insights	Pay-as-you-go	15 GB log ingestion/month	\$30
Log Analytics Workspace	Per GB	Retention: 30 days	\$25
Container Insights	Add-on	AKS pod/node metrics	\$20
Observability subtotal			~\$75/mo

Option 2 — Infrastructure Subtotal

Category	Est. USD/mo
AKS Cluster	\$570
Data & Storage	\$614
Networking & Security	\$156
Container Registry	\$20
Observability	\$75
Total Infrastructure	~\$1,435/mo

AI/LLM Services (Variable — same as Option 1)

Service	Low Usage	Medium Usage	High Usage
Azure OpenAI (all models)	\$87	\$278	\$685

Service	Low Usage	Medium Usage	High Usage
Mistral AI OCR	\$10	\$40	\$120
AI/LLM Subtotal	~\$97/mo	~\$318/mo	~\$805/mo

Option 2 — Total Monthly Cost

Scenario	Infrastructure	AI/LLM	Total	Total (AED)
Low usage	\$1,435	\$97	~\$1,532/mo	~5,622 AED/mo
Medium usage	\$1,435	\$318	~\$1,753/mo	~6,434 AED/mo
High usage	\$1,435	\$805	~\$2,240/mo	~8,221 AED/mo

Option 2 — Production Capabilities

Capability	Detail
Auto-scaling	Horizontal Pod Autoscaler (HPA) on FastAPI pods; Cluster Autoscaler adds/removes D4s_v3 nodes as needed
Zero-downtime deployments	Rolling updates with <code>maxUnavailable: 0</code> , <code>maxSurge: 1</code> ; readiness probes gate traffic
Database failover	Zone-redundant standby: automatic failover in <120 seconds; no manual intervention
WAF protection	OWASP Core Rule Set 3.2 blocks SQLi, XSS, and common API attacks
Email Monitor resilience	Kubernetes restart policy <code>OnFailure</code> ; liveness probe restarts stuck IMAP connections
Secrets management	Azure Key Vault CSI driver mounts secrets directly into pods; no environment variable exposure
Observability	Container Insights provides pod CPU/memory, restart counts; App Insights traces FastAPI requests; Langfuse traces LLM calls
Availability SLA	99.95% AKS + 99.99% PostgreSQL HA = ~99.94% composite SLA
pgvector performance	16 GB RAM per node; pgvector HNSW index fits comfortably for 500k+ embeddings

Side-by-Side Comparison

Dimension	Option 1 (App Service)	Option 2 (AKS)
Monthly infra cost	~\$173	~\$1,435

Dimension	Option 1 (App Service)	Option 2 (AKS)
Total (medium usage)	~\$491/mo	~\$1,753/mo
Setup complexity	Low — PaaS, GitHub Actions deploy	High — Helm charts, manifests, ACR pipelines
Deployment time (initial)	1–2 days	1–2 weeks
Auto-scaling	None (manual only)	Yes — pod + node autoscaling
Zero-downtime deploy	No (unless +deployment slot)	Yes — rolling updates
Database HA	No standby, ~99.9% SLA	Zone-redundant standby, ~99.99% SLA
Failover	Manual restart	Automatic <120s
WAF / DDoS protection	None	Application Gateway WAF v2
Concurrent users (comfortable)	10–30	100–500+
Agent orchestration headroom	Limited — 12 agents compete for single instance	Strong — pods scale independently
Email monitor reliability	WebJob shares App Service resources	Isolated pod with dedicated restart policy
Observability	Basic App Insights	Full Container Insights + Log Analytics
Secrets management	App Settings (env vars)	Key Vault CSI driver (no plain-text env)
Recommended for	Dev, staging, pilot, <30 users	Production, >30 users, SLA commitment

Cost Optimization Notes

Option 1 Optimizations

- Use **Azure Reserved Instances (1-year)** for App Service and PostgreSQL to save ~30%: reduces infra to ~\$120/mo.
- Use **Free tier Static Web Apps** if custom auth via JWT is sufficient (saves \$9/mo).
- Enable **pgvector HNSW index** over IVFFlat for better query performance without upgrading DB tier.

Option 2 Optimizations

- Use **Spot node pools** for the app node pool (non-system workloads): up to 60–80% savings on compute (~\$165/mo instead of \$414/mo for app nodes), but requires pods to tolerate eviction.

- Use **Azure Reserved Instances (1-year)** on system nodes and PostgreSQL: saves ~30% on those line items.
- Use **AKS Start/Stop** for dev/staging clusters to avoid weekend costs.
- Consider **Azure Container Apps** as a middle path: serverless containers with scale-to-zero, ~\$300–600/mo infrastructure, simpler than AKS but more capable than App Service.

Shared Optimizations (both options)

- Enable **Azure OpenAI prompt caching** where supported to reduce token costs.
- Use **GPT-4.1-mini** for all non-critical agent steps (matcher, simple routing) — already partially implemented.
- Set a **PRICE_THRESHOLD** tuned to minimize unnecessary GPT-4.1 calls.
- Monitor Langfuse token usage per agent to identify cost spikes.

Recommended Migration Path

Phase 1 (Now) Option 1	Phase 2 (Scale) Option 1+	Phase 3 (Enterprise) Option 2
App Service P1v3 → PostgreSQL B2ms → Static Web Apps → ~\$173 infra/mo	App Service P2v3 → PostgreSQL D2s_v3 → Static Web Apps → ~\$280 infra/mo	AKS (D4s_v3 nodes) PostgreSQL D4s_v3 + HA CDN + App Gateway WAF ~\$1,435 infra/mo

The trigger to move from Option 1 → Option 2 is typically:

- User count exceeds 30 concurrent users, **OR**
- A database/App Service outage causes a business impact, **OR**
- An enterprise SLA or compliance requirement (ISO 27001, etc.) is mandated.